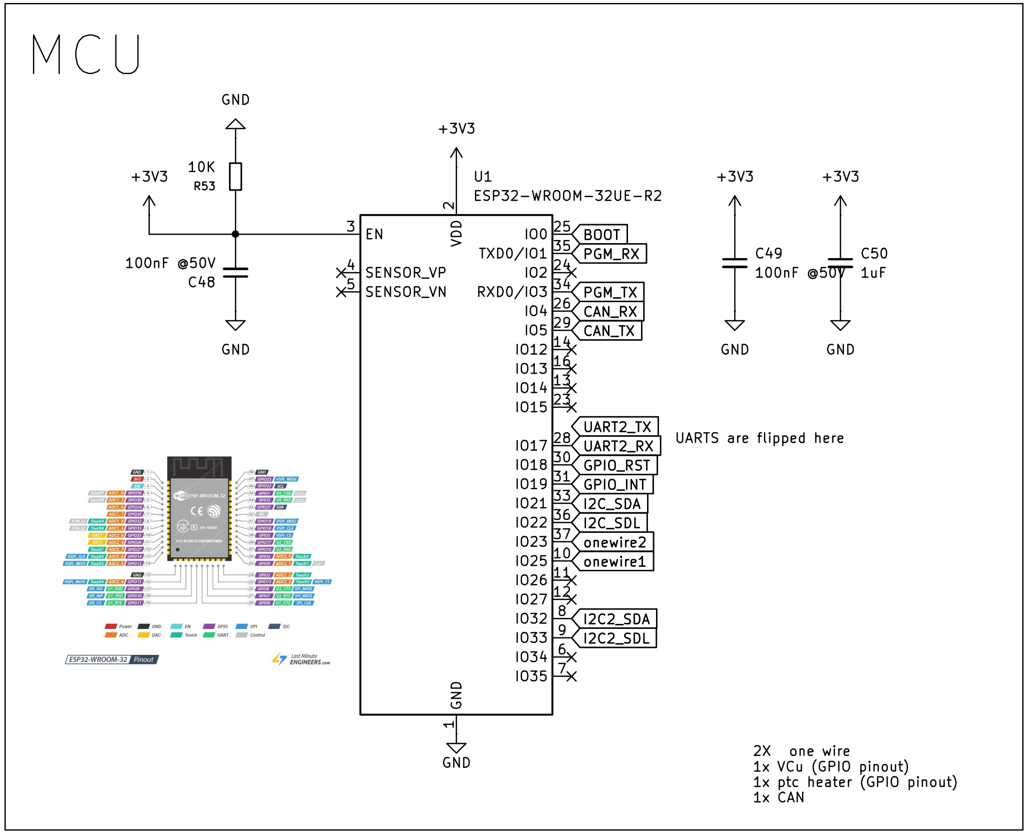
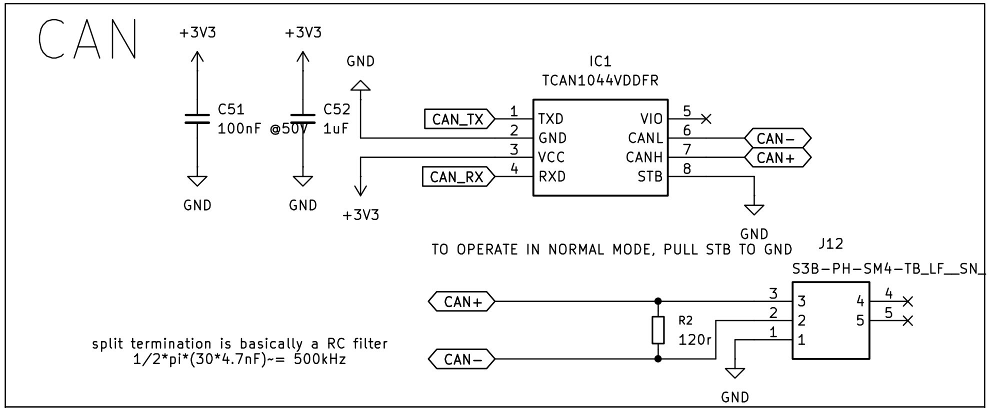
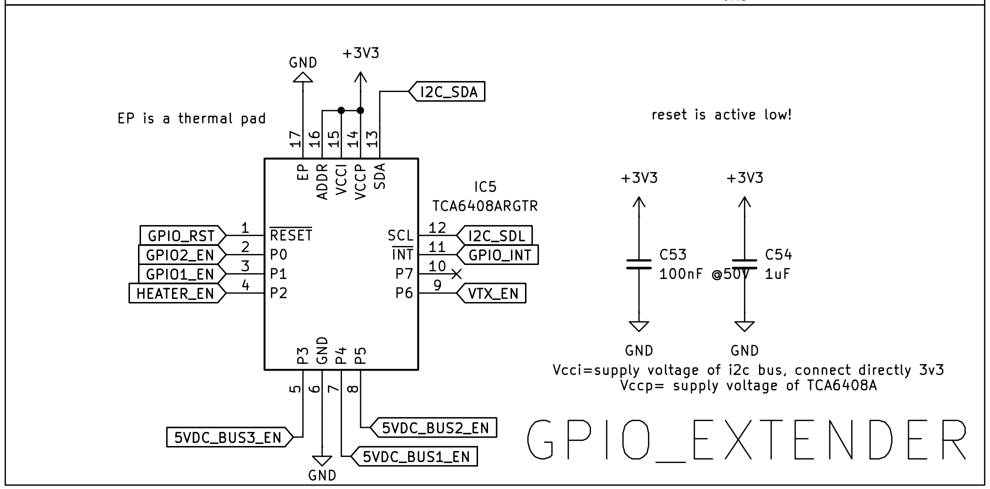
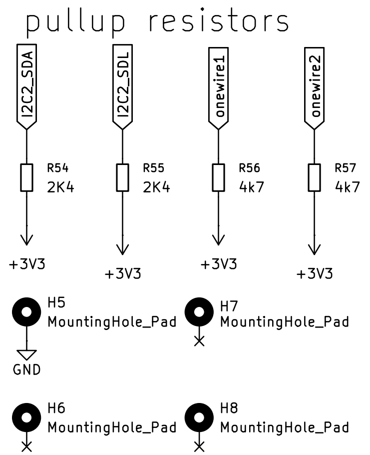
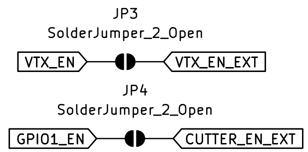
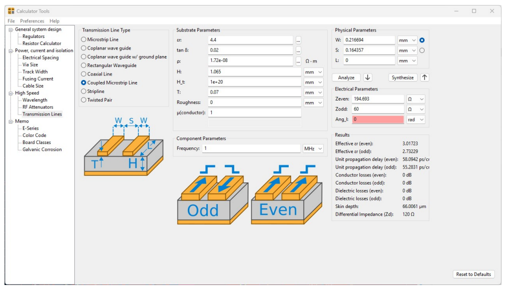








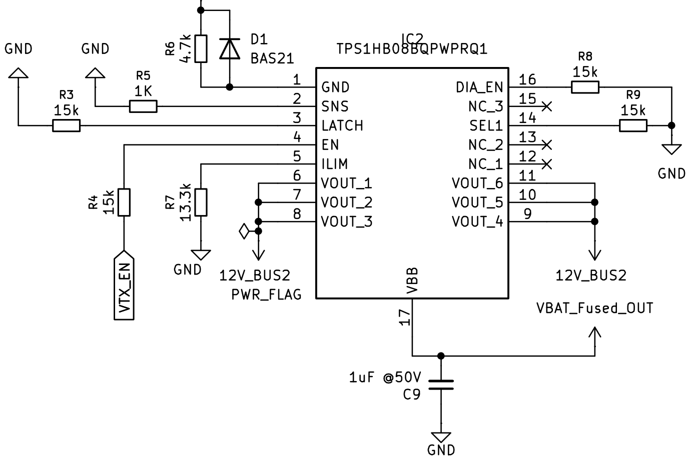




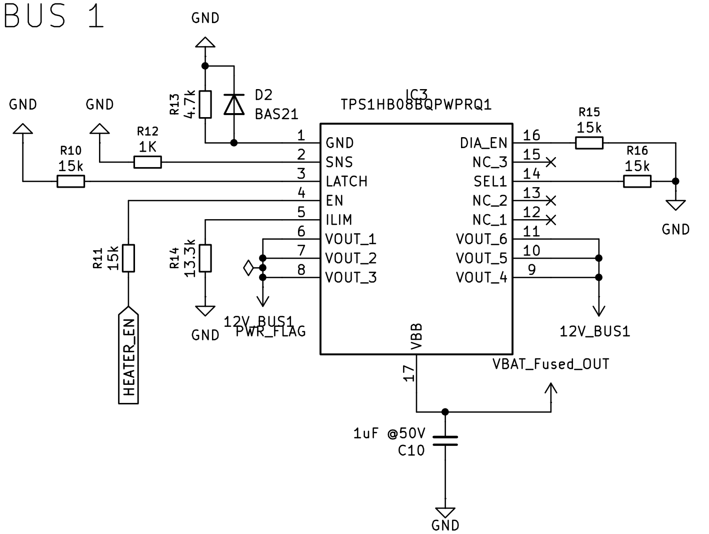
2) JLC04162H-7628 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.07mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	
Core	Core	1.065mm	1.1mm H/HOZ with copper
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.07mm	

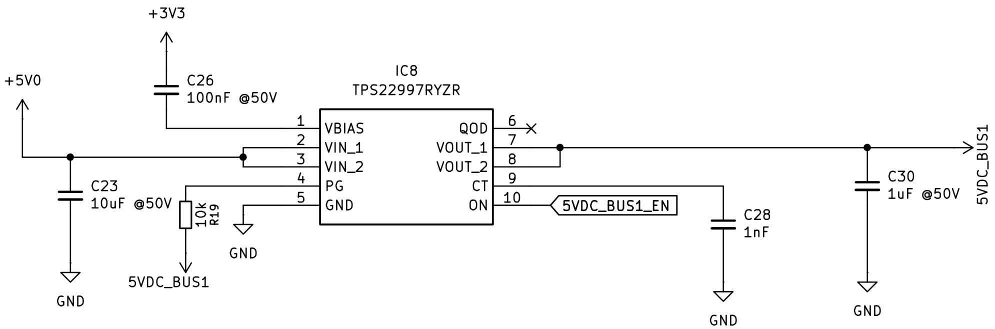
BUS 1



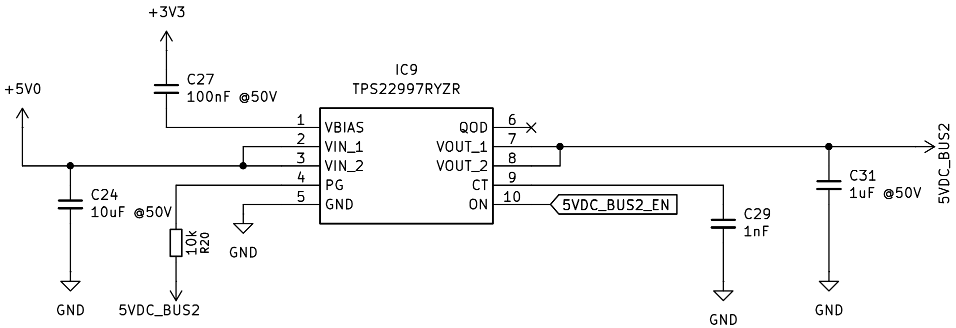
BUS 1



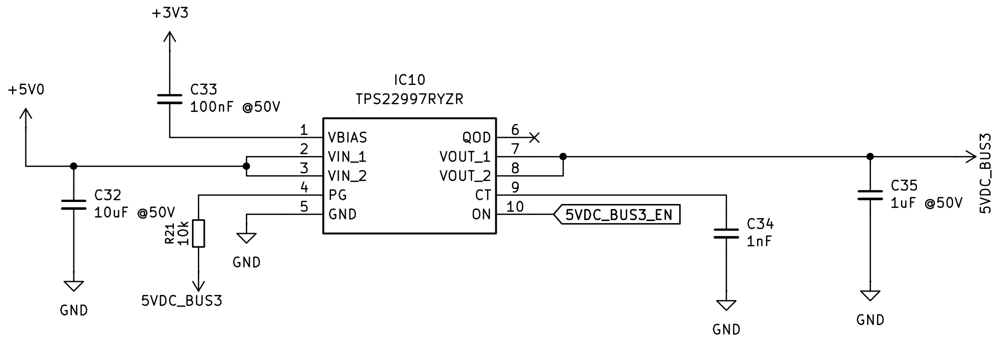
5VDC BUS 1



5VDC BUS 2



5VDC BUS 1



QOD pin shorted to VOUT pin. Using this method, the discharge rate after the switch becomes disabled is controlled with the value of the internal resistance RQOD. The value of this resistance is listed in the Electrical Characteristics table.

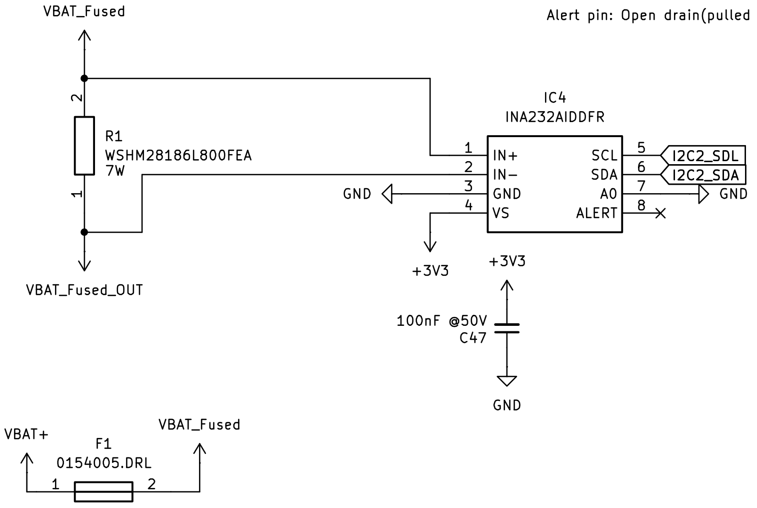
PG:
The signal is an active high and open drain output which can be connected to a voltage source through an external pullup resistor, RPU. This voltage source can be VOUT from the TPS22997 device or another external voltage.

CURRENT SENSE_ VBAT

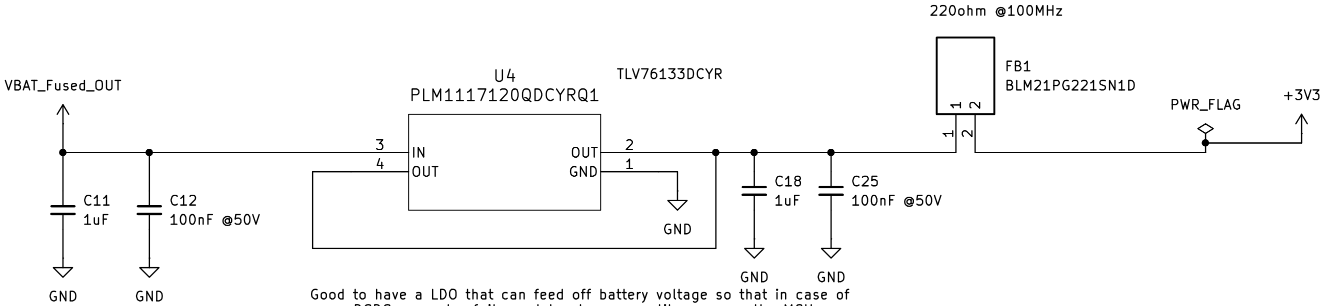
!Device Address!
AO connected to VS:1000000
AO connected to GND:1000001

Alert pin: Open drain(pulled to ground) output, effective when limits exceeded or conversion ready

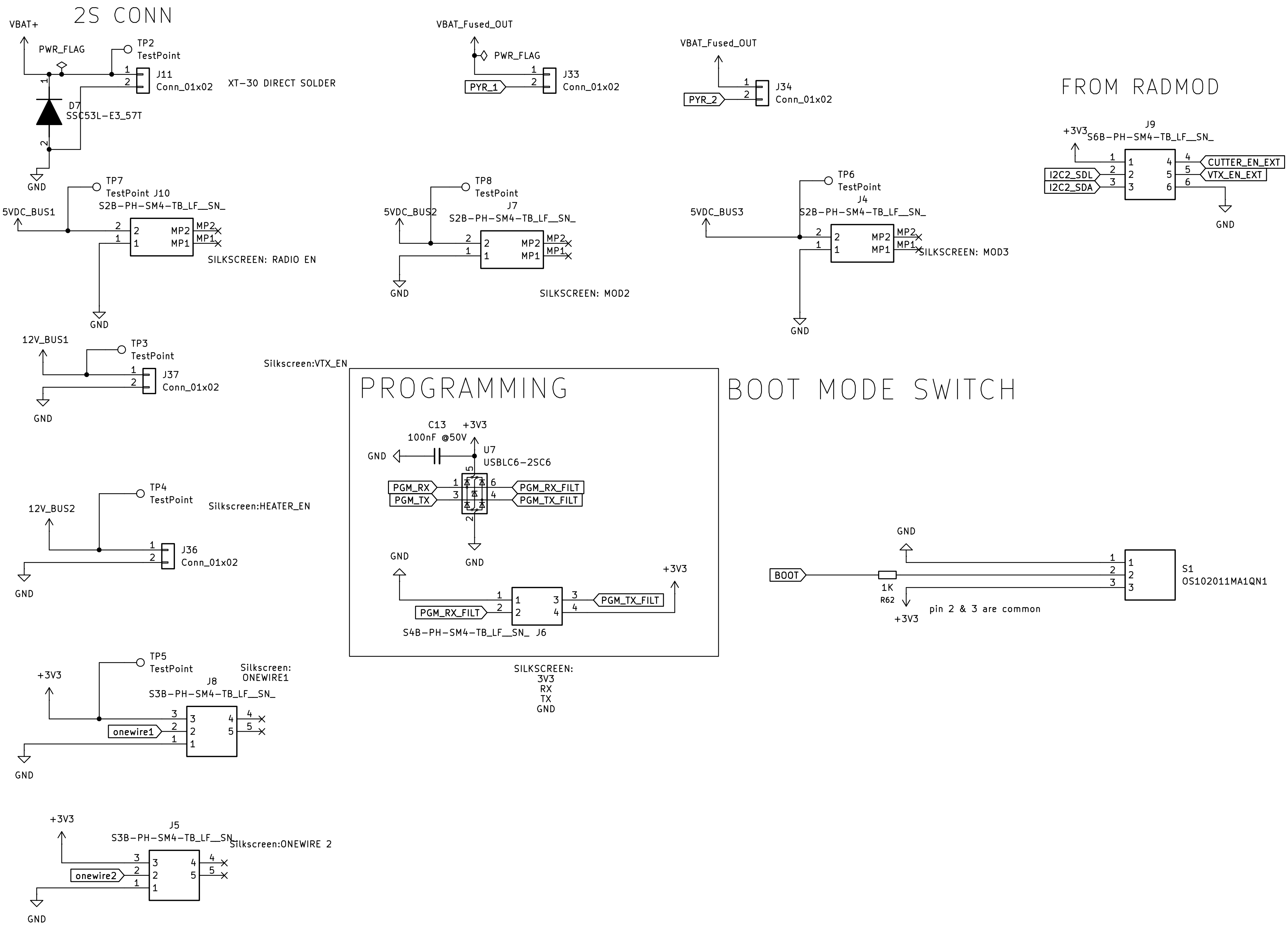
Vbat current sense:
Rshunt< Vsenese/I max
Vsense= ±81.92 mV (ADCRANGE = 0), Imax=12[Ilim of all load switches], Rshunt=6.827mohm
Power= 6.8e−03*12^2=0.9792

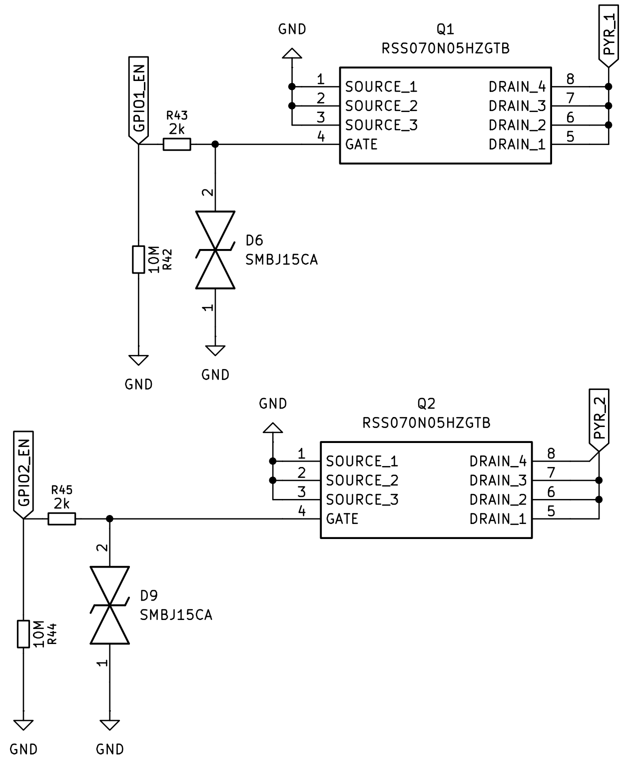


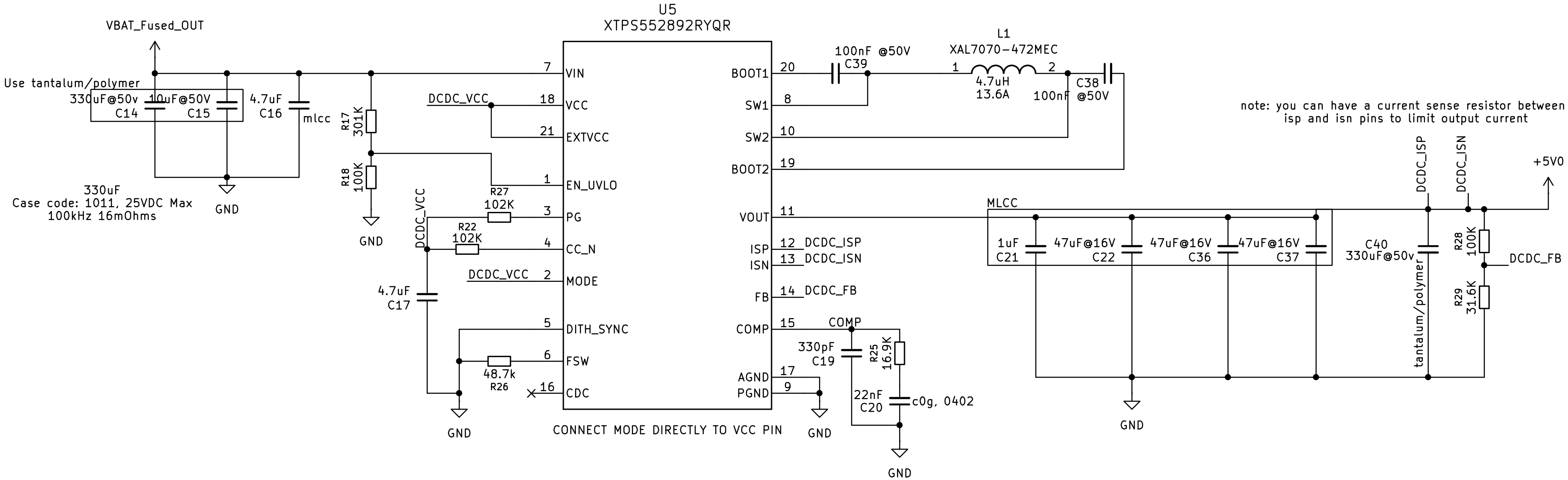
Fixed Output at 3.3V!



Good to have a LDO that can feed off battery voltage so that in case of
DCDC covnerter failure at least we can stil power on the MCU


2S CONN
VBAT+
PWR_FLAG
TP2
TestPoint
J11
Conn_01x02
XT-30 DIRECT SOLDER
D7
SSC53L-E3_57T
GND
VBAT_Fused_OUT
PWR_FLAG
J33
Conn_01x02
PYR_1
VBAT_Fused_OUT
J34
Conn_01x02
PYR_2
FROM RADMOD
J9
+3V3
S6B-PH-SM4-TB_LF__SN_
CUTTER_EN_EXT
I2C2_SDL
VTX_EN_EXT
I2C2_SDA
GND
TP7
TestPoint J10
S2B-PH-SM4-TB_LF__SN_
5VDC_BUS1
SILKSCREEN: RADIO EN
GND
TP8
TestPoint
J7
5VDC_BUS2
S2B-PH-SM4-TB_LF__SN_
GND
SILKSCREEN: MOD2
TP6
TestPoint
J4
5VDC_BUS3
S2B-PH-SM4-TB_LF__SN_
SILKSCREEN: MOD3
GND
12V_BUS1
TP3
TestPoint
J37
Conn_01x02
GND
Silkscreen:VTX_EN
PROGRAMMING
BOOT MODE SWITCH
C13
100nF @50V
+3V3
GND
U7
USBLC6-2SC6
PGM_RX
PGM_TX
PGM_RX_FILT
PGM_TX_FILT
GND
GND
+3V3
PGM_TX_FILT
PGM_RX_FILT
S4B-PH-SM4-TB_LF__SN_ J6
SILKSCREEN:
3V3
RX
TX
GND
TP4
TestPoint
12V_BUS2
Silkscreen:HEATER_EN
J36
Conn_01x02
GND
GND
BOOT
1K
R62
+3V3
pin 2 & 3 are common
S1
OS102011MA1QN1
TP5
TestPoint
+3V3
Silkscreen:
ONEWIRE1
J8
S3B-PH-SM4-TB_LF__SN_
onewire1
GND
+3V3
J5
S3B-PH-SM4-TB_LF__SN_
Silkscreen:ONEWIRE 2
onewire2
GND


GND
Q1
RSS070N05HZGTB
PYR_1
GPIO1_EN
R43
2k
1
2
3
4
SOURCE_1
SOURCE_2
SOURCE_3
GATE
DRAIN_4
DRAIN_3
DRAIN_2
DRAIN_1
8
7
6
5
2
D6
SMBJ15CA
1
10M
R42
GND
GND
GND
Q2
RSS070N05HZGTB
PYR_2
GPIO2_EN
R45
2k
1
2
3
4
SOURCE_1
SOURCE_2
SOURCE_3
GATE
DRAIN_4
DRAIN_3
DRAIN_2
DRAIN_1
8
7
6
5
2
D9
SMBJ15CA
1
10M
R44
GND
GND


U5
XTPS552892RYQR
VBAT_Fused_OUT
Use tantalum/polymer
330uF@50v
10uF@50V
4.7uF
C14
C15
C16
mlcc
330uF
Case code: 1011, 25VDC Max
100kHz 16mOhms
GND
R17
301K
R18
100K
DCDC_VCC
R27
102K
R22
102K
4.7uF
C17
48.7k
R26
VIN
VCC
EXTVCC
EN_UVLO
PG
CC_N
MODE
DITH_SYNC
FSW
CDC
AGND
PGND
BOOT1
SW1
SW2
BOOT2
VOUT
ISP
ISN
FB
COMP
DCDC_ISP
DCDC_ISN
DCDC_FB
CONNECT MODE DIRECTLY TO VCC PIN
100nF @50V
C39
L1
XAL7070-472MEC
4.7uH
13.6A
C38
100nF @50V
330pF
C19
R25
16.9K
22nF
C20
c0g, 0402
note: you can have a current sense resistor between
isp and isn pins to limit output current
MLCC
1uF
C21
47uF@16V
C22
C36
C37
C40
330uF@50v
tantalum/polymer
R28
100K
R29
31.6K
+5V0